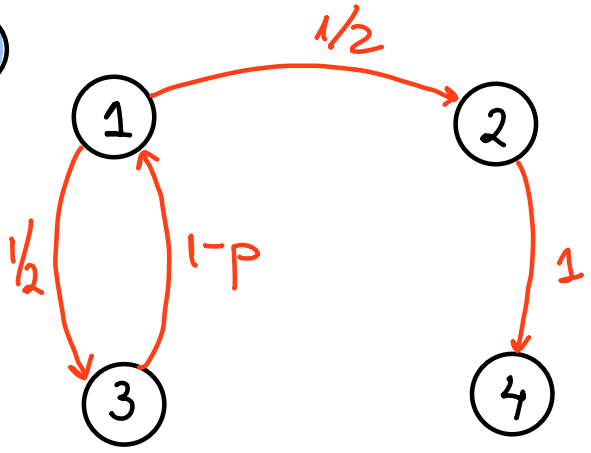
Problemă Faceți primii pași pentru Givens/G-S/Haesholder pe mat. $\begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$

ex



$$P = \begin{bmatrix} 0 & 0,5 & 0,5 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

1/4 1/4 1/4 1/4

$$M = \alpha \cdot P + (1-\alpha) \cdot \frac{1}{n} \cdot V \cdot e^t$$

$$\rho(M) = 1$$

$$M\Pi = \Pi$$

Page Rank

Valori și vectori proprii

→ vectorii proprii arată un comportament (importanța unui lucru asupra tuturor celorlalte)

Obs

matrici asemenea $A \sim B \Leftrightarrow \exists P$ a.î. $A = P^{-1}BP$

prin urmare $P_\lambda(A) = P_\lambda(B)$

$A \in \mathbb{R}^{n \times n}$ $P_\lambda(A) = \det(A - \lambda I_n) = 0 \Rightarrow$

$\Rightarrow \lambda(A) = \{\lambda_i \in \mathbb{C} \mid P(\lambda_i) = 0;\ i \in 1:n\}$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \qquad |z - a_{ii}| \leq \sum_{\substack{j=0 \\ j \neq i}}^{n} a_{ij}$$

Propoziție $A \in \mathbb{R}^{n \times n}$, $\lambda(A)$

$$x(A) = \{ x_i \in \mathbb{R}^n \mid Ax_i = \lambda_i x_i,\ \lambda_i \in \lambda(A);\ i = \overline{1:n} \}$$

cu proprietățile

a) $\lambda(A - \mu I_n) = \{ \lambda_i - \mu \mid \lambda_i \in \lambda(A);\ i = \overline{1:n} \}$

b) $\lambda(A^k) = \{ \lambda_i^k \mid \lambda_i \in (A);\ i = \overline{1:n};\ k \in \mathbb{N} \}$

c) $\lambda((A - \mu I)^{-1}) = \left\{ \frac{1}{\lambda_i - \mu} \mid \lambda_i \in \lambda(A);\ i = \overline{1:n} \right\}$

$\longrightarrow$ se păstrează semnificația vectorilor proprii

$$\left.\begin{aligned} (A - \mu I_n)x_i &= (\lambda_i - \mu)x_i \\ A^k x_i &= \lambda_i^k x_i \\ (A - \mu I_n)^{-1} x_i &= \frac{1}{\lambda_i - \mu} x_i \end{aligned}\right| \quad x_i \in x(A)$$

Metode numerice

a) MPD (Metoda puterii directe)

$A \in \mathbb{R}^{n \times n}$ $\quad |\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_n|$

λ_1 ↑ val. prop. dominantă

Fie $y \in \mathbb{R}^n$ atunci $y;\ Ay;\ A^2y \dots A^k y \dots \longrightarrow x_1$

$$\Rightarrow \lambda_1 = \frac{x_1^T A x_1}{x_1^T x_1}$$

P.1: $y^{(0)} = y$

P.2: pentru $k = 1, 2, \dots$ $\quad y^{(0)} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$

$$z = A \cdot y^{(k-1)}$$

$$y^{(k)} = z / \|z\|_2$$

$$\lambda^{(k)} = y^{(k)T} \cdot A \cdot y^{(k)}$$

Ne oprim: $\|y^{(k)} - y^{(k-1)}\|_2 < \varepsilon$

sau nr. maxim de pași

$$\|(A - \lambda^{(k)} I_n) z\| < \varepsilon$$

$$y^{(0)} = \sum_{i=1}^{n} \alpha_i x_i \quad \alpha_i \in \mathbb{R}$$

! $\alpha_1 \neq 0$

$$y^{(k)} = \rho_k A^k y^{(0)}$$

$$y^{(k)} = \rho_k A^k \sum_{i-1}^{n} \alpha_i x_i = \rho_k \sum_{i=1}^{n} \alpha_i A^k \cdot x_i = \rho_k \sum_{i=1}^{n} \alpha_i \underbrace{\lambda_i^k x_i}_{\dots} \cdot x_i$$

$$y^{(k)} = \rho_k \alpha_1 \lambda_1^k \left[x_1 + \sum_{i=2}^{n} \frac{\alpha_i}{\alpha_1} \left(\frac{\lambda_i}{\lambda_1} \right)^k \cdot x_i \right]$$

$\left(\frac{\lambda_i}{\lambda_1} \right)^k \to 0$

$k \to \infty \qquad y^{(k)} \longrightarrow \rho_k \cdot \alpha_1 \lambda_1^k \cdot x_1$

$\rho_k \cdot \alpha_1 \lambda_1^k$ finit

! Afla valoare proprie dominanta

! Viteza de convergenta $\sim \frac{|\lambda_2|}{|\lambda_1|}$

Dem

$$v = a_1 x_1 + a_2 x_2 + \dots + a_n x_n$$

$$Av = a_1 Ax_1 + a_2 Ax_2 + \dots + a_n Ax_n$$

$$= a_1 \lambda_1 x_1 + a_2 \lambda_2 x_2 + \dots + a_n \lambda_n x_n$$

$$= \lambda_1 \left(a_1 x_1 + a_2 \frac{\lambda_2}{\lambda_1} x_2 + \dots + a_n \frac{\lambda_n}{\lambda_1} x_n \right)$$

v_1

$$Av_1 = \lambda_1 \left(a_1 x_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^2 x_2 + \dots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^2 x_n \right)$$

$$\vdots$$

$$Av_k = \lambda_1 \left(a_1 x_1 + a_2 \left(\frac{\lambda_2}{\lambda_1} \right)^{k+1} x_2 + \dots + a_n \left(\frac{\lambda_n}{\lambda_1} \right)^{k+1} x_n \right)$$

$k \to \infty \Rightarrow$ $\left(\frac{\lambda_2}{\lambda_1} \right)^{k+1} \to 0$

b) MPi (Metoda puterii inverse)

Dacă $\mu \to$ aproximare λ_i $i \neq 1$ atunci $\frac{1}{\lambda_i - \mu}$ este val. prop. dominantă pt. $(A - \mu I_n)^{-1}$

P1: $y^{(0)} = y$ $\qquad y^{(0)} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$

P2: pentru $i = 1, 2, \ldots$

rezolvăm $(A - \mu I_n) \cdot z = y^{(k-1)}$

$y^{(k)} = z / \|z\|_2$

$\lambda^{(k)} = y^{(k)T} A y^{(k)}$

Optimizare: pentru $i = 1, 2, \ldots$

rezolvăm $(A - \mu I_n) \cdot z = y^{(k-1)}$

$y^{(k)} = z / \|z\|_2$

$\mu = y^{(k)T} A y^{(k)}$

$\lambda = \mu$

⚠ Tot descoperă un λ_i care nu e λ_1, dar nu se știe care

Valori proprii pentru $(A - \mu I)^{-1} \to x$ și λ

c) **Deflatia valorilor proprii** → doar pt. matrici simetrice

$A \to (\lambda_1, x_1)$

Formăm $B = (I_n - x_1 y^T) A$ și $y^T x_1 = 1$

! posibilă alegere $y = \alpha x_1$

$\alpha x_1^T \cdot x_1 = 1 \Rightarrow \alpha = \frac{1}{\|x_1\|^2}$ $y = \frac{x_2}{\|x\|_1^2}$

Demo $B \cdot x_1 = (I_n - x_1 y^T) A \cdot x_1 = \lambda_1 (I_n - x_1 y^T) x_1$

$= \lambda_1 (x_1 - x_1 \underbrace{y^T x_1}_{1}) = \lambda_1 (x_1 - x_1) = 0$ ⇒

Alegem $\lambda_1 = 0$ pt. că avem nevoie de x_1

$\Rightarrow B x_1 = 0 \cdot x_1$ intuitiv $\lambda(B) = \{0, \lambda_2, \dots \lambda_n\}$

$x(B) = \{x_1, z_2, \dots z_n\}$

$B \cdot x_i = \lambda_i (\underbrace{x_i - y^T x_i x_1}_{z_i}) = \lambda_i z_i$

$B \cdot z_i = B(x_i - y^T x_i x_1) = Bx_i - (y^T x_i) \underbrace{Bx_1}_{0} = Bx_i = \lambda_i z_i$

→ aplicăm met. puterii pe A → obținem x_1, λ_1

→ formăm $B = (I_n - x_1 y^T) A$

→ aplicăm met. puterii pe B → obținem x_2, λ_2

(B fără prima linie + col)